

Stoichiometry/Solutions

- How many atoms are there in 0.300 mol of potassium nitrate, KNO_3 ($M = 101.11$)?
(A) 1.79×10^{21} (B) 1.81×10^{23}
(C) 5.42×10^{23} (D) 9.03×10^{23}
- A student isolates 22.0 g of ethyl bromide, $\text{C}_2\text{H}_5\text{Br}$ ($M = 108.96$), from the reaction of 34.0 g of ethyl alcohol ($M = 46.07$) with 59.0 g of phosphorus tribromide ($M = 270.67$) according to the balanced reaction below. What is the percent yield of the reaction?
 $3 \text{C}_2\text{H}_5\text{OH}(l) + \text{PBr}_3(l) \rightarrow 3 \text{C}_2\text{H}_5\text{Br}(l) + \text{H}_3\text{PO}_3(s)$
(A) 27.4% (B) 30.9% (C) 64.7% (D) 92.6%
- A 25.00 mL sample of aqueous HCl is titrated with 0.150 M NaOH , which requires adding 18.75 mL of the NaOH solution to reach the equivalence point. What is the concentration of chloride ion in the final solution?
(A) 0.0643 M (B) 0.0750 M
(C) 0.112 M (D) 0.200 M
- A colorless liquid hydrocarbon contains carbon and hydrogen in a 6.0 : 1 ratio by mass. Which is its molecular formula?
(A) CH (B) CH_2
(C) C_5H_{10} (D) C_6H_6
- Mixing equal volumes of 0.15 M $\text{Ba}(\text{NO}_3)_2$ and 0.20 M NaF causes the precipitation of insoluble barium fluoride. In which are the concentrations of ions in the supernatant listed in increasing order?
(A) $[\text{Ba}^{2+}] < [\text{F}^-] < [\text{NO}_3^-] < [\text{Na}^+]$
(B) $[\text{Ba}^{2+}] < [\text{F}^-] < [\text{Na}^+] < [\text{NO}_3^-]$
(C) $[\text{F}^-] < [\text{Ba}^{2+}] < [\text{NO}_3^-] < [\text{Na}^+]$
(D) $[\text{F}^-] < [\text{Ba}^{2+}] < [\text{Na}^+] < [\text{NO}_3^-]$
- A solution of which compound, at a concentration of 25 g L^{-1} , has the lowest freezing point?
(A) NaBrO_3 (B) KBrO_3
(C) Na_2SeO_4 (D) K_2SeO_4
- How many nitrogen atoms are present in 486 g of N_2O_5 ?
(A) 1.35×10^{24} (B) 2.71×10^{24}
(C) 5.42×10^{24} (D) 1.90×10^{25}
- What is the empirical formula of a compound that is 50% S and 50% O by mass?
(A) SO (B) S_2O
(C) SO_2 (D) S_2O_3
- How much chlorine is required to completely consume 13.5 g of aluminum to form aluminum chloride?
(A) 13.5 g (B) 26.6 g
(C) 40.5 g (D) 53.2 g
- A solution of 0.500 L of 0.150 M KI is mixed with 0.250 L of 0.200 M CaI_2 . What is the concentration of iodide in the final solution?
(A) 0.167 M (B) 0.175 M
(C) 0.233 M (D) 0.275 M
- The only source of copper in an ore sample is the mineral malachite, $\text{Cu}_2\text{CO}_3(\text{OH})_2$ ($M = 221.1$). A 1.00 kg sample of this ore is smelted to obtain 47.0 g of metallic copper. Assuming a complete recovery of the copper, what percentage by mass of the ore is malachite?
(A) 4.70% (B) 8.18%
(C) 16.4% (D) 32.8%
- Which aqueous solution has the highest boiling point when 10.00 g of the substance is dissolved in 100.0 g of water?
(A) H_2SO_4 (B) Li_2SO_4
(C) Na_2SO_4 (D) MgSO_4
- How many moles of ions are in 75.0 mL of a 1.20 M aqueous solution of sodium sulfate?
(A) 0.090 mol (B) 0.120 mol
(C) 0.180 mol (D) 0.270 mol
- When 3.00 g of $\text{NH}_3(g)$ ($M = 17.03$) reacts with 18.0 g of $\text{HCl}(g)$ ($M = 36.46$), what is the theoretical yield of $\text{NH}_4\text{Cl}(s)$?
(A) 6.42 g (B) 8.41 g
(C) 9.42 g (D) 21.0 g
- How many fluorine atoms are in 135 g of CF_2Cl_2 ?
(A) 3.36×10^{23} (B) 6.72×10^{23}
(C) 1.34×10^{24} (D) 4.28×10^{24}

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16. A 2024 US penny (which is a mixture of zinc and copper) is dissolved in dilute nitric acid, forming aqueous Cu^{2+} and Zn^{2+} ions. The resulting solution is treated with an excess of iodide ion, causing the Cu^{2+} to react to form $\text{CuI}(s)$ and $\text{I}_3^-(aq)$. Subsequent titration with thiosulfate reveals that 4.93×10^{-4} mol of $\text{I}_3^-(aq)$ was formed. What is the mass of copper in the penny?
- (A) 31.3 mg (B) 62.7 mg
(C) 125 mg (D) 188 mg
17. 1.22 g of benzoic acid ($\text{C}_7\text{H}_6\text{O}_2$, $M = 122.12$) is dissolved in water to a total volume of 150.0 mL. A 10.0 mL sample of this solution is then withdrawn and diluted to 500.0 mL. What is the concentration of benzoic acid in the final solution?
- (A) 0.00133 M (B) 0.00266 M
(C) 0.0200 M (D) 0.122 M
18. A 0.75 *m* aqueous solution of which substance has the lowest freezing point?
- (A) HF (B) HI
(C) H_3PO_4 (D) CH_3COOH
19. A 1.00 g sample of which compound will produce the greatest amount of carbon dioxide after complete combustion with excess oxygen?
- (A) CH_4 (B) C_3H_8 (C) C_6H_6 (D) C_8H_{18}
20. How many chlorine atoms are in 1.00 mL of CCl_4 ($M = 153.8$, density = 1.59 g mL^{-1})?
- (A) 1.56×10^{21} (B) 6.23×10^{21}
(C) 1.57×10^{22} (D) 2.49×10^{22}
21. What is the molality of Na^+ in a solution containing 3.00 g NaCl ($M = 58.4$), 9.00 g glucose ($M = 180.0$), and 168 g H_2O ($M = 18.0$)?
- (A) $5.50 \times 10^{-3} m$ (B) 0.285 *m*
(C) 0.306 *m* (D) 0.777 *m*
22. 25.0 mL of aqueous 0.20 M HCl , 10.0 mL of 0.25 M aqueous HNO_3 , and 10.0 mL of aqueous 0.30 M $\text{Ba}(\text{OH})_2$ are mixed to give a homogeneous solution. In which are the final concentrations of species ranked in order from highest to lowest?
- (A) $[\text{Cl}^-] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{H}^+] > [\text{OH}^-]$
(B) $[\text{H}^+] > [\text{OH}^-] > [\text{Cl}^-] > [\text{Ba}^{2+}] > [\text{NO}_3^-]$
(C) $[\text{Cl}^-] > [\text{H}^+] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{OH}^-]$
(D) $[\text{OH}^-] > [\text{H}^+] > [\text{Ba}^{2+}] > [\text{NO}_3^-] > [\text{Cl}^-]$
23. Acetanilide ($\text{C}_6\text{H}_5\text{NHCOCH}_3$, $M = 135.2$) can be prepared from aniline ($\text{C}_6\text{H}_5\text{NH}_2$, $M = 93.1$) and acetic anhydride ($[\text{CH}_3\text{CO}]_2\text{O}$, $M = 102.1$) according to the balanced equation below.
- $$\text{C}_6\text{H}_5\text{NH}_2 + (\text{CH}_3\text{CO})_2\text{O} \rightarrow \text{C}_6\text{H}_5\text{NHCOCH}_3 + \text{CH}_3\text{COOH}$$
- If 10.0 g acetanilide is isolated from the reaction of 10.0 g aniline with 10.0 g acetic anhydride, what is the percent yield of the reaction?
- (A) 50.0% (B) 68.9% (C) 69.3% (D) 75.5%
24. 10.0 g of a metal chloride salt is dissolved in 90.0 g H_2O . This solution has a vapor pressure that is 8.6% lower than that of pure water at the same temperature. What is the identity of the metal?
- (A) Li (B) Na (C) Mg (D) Ca
25. A 1.00-g sample of which compound will produce the greatest amount of carbon dioxide after complete combustion with excess oxygen?
- (A) CH_4 (B) C_3H_6 (C) C_6H_{14} (D) C_8H_{18}
26. Rhenium (Re) consists of two stable isotopes, ^{185}Re and ^{187}Re . The average atomic mass of Re is 186.2 amu. What is the mole percent of ^{185}Re in rhenium?
- (A) 40% (B) 50%
(C) 60% (D) 62%
27. A 10.00 g sample of a soluble barium salt is treated with an excess of sodium sulfate to precipitate 8.93 g BaSO_4 ($M = 233.4$). Which barium salt is it?
- (A) BaCl_2 ($M = 208.2$)
(B) $\text{Ba}(\text{O}_2\text{CH})_2$ ($M = 227.3$)
(C) $\text{Ba}(\text{NO}_3)_2$ ($M = 261.3$)
(D) BaBr_2 ($M = 297.1$)

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28. How much $\text{Sr}(\text{OH})_2 \cdot 8 \text{H}_2\text{O}$ ($M = 265.76$) is needed to prepare 150.0 mL of solution in which $[\text{OH}^-] = 0.100 \text{ M}$?

- (A) 1.99 g (B) 3.98 g
(C) 5.98 g (D) 7.98 g

29. Which procedure produces a 0.2 M solution of Na_2SO_4 ?

- (A) Mixing 500 mL of 0.4 M NaOH with 500 mL of 0.2 M Na_2SO_4
(B) Mixing 300 mL of 1.2 M NaOH with 600 mL of 0.3 M H_2SO_4
(C) Dissolving 0.2 mol solid Na_2SO_4 in 500 mL water
(D) Diluting 400 mL of 0.3 M Na_2SO_4 to a final volume of 1.0 L

30. 10.0 g of an alkali metal chloride salt is dissolved in 90.0 g H_2O . This solution has a vapor pressure that is 3.2% lower than that of pure water at the same temperature. What is the salt?

- (A) LiCl (B) NaCl (C) KCl (D) RbCl

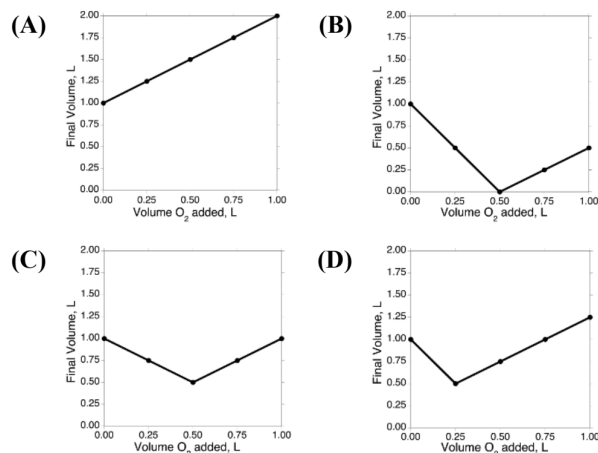
31. Which potassium salt is 28.9% chlorine by mass?

- (A) KCl (B) KClO (C) KClO_2 (D) KClO_3

32. How many sulfur atoms are in 3.00 g of iron pyrite, FeS_2 ($M = 120.0$)?

- (A) 7.53×10^{21} (B) 1.51×10^{22}
(C) 3.01×10^{22} (D) 6.02×10^{23}

33. A gas mixture contains 50 mol% H_2 and 50 mol% He. 1.00-L samples of this gas mixture are mixed with variable volumes of O_2 (at 0 °C and 1 atm). A spark is introduced to allow the mixture to undergo complete combustion. The final volume is measured at 0 °C and 1 atm. Which graph best depicts the final volume as a function of the volume of added O_2 ?



34. What is the concentration of nitrate ions in 40.0 mL of a 0.25 M solution of copper(II) nitrate?

- (A) 0.010 M (B) 0.020 M
(C) 0.050 M (D) 0.50 M

35. 1.000 g of a transition metal carbonate, MCO_3 , is heated to produce the metal oxide and 0.383 g of carbon dioxide ($M = 44.01$). What is the identity of the metal M?

- (A) Mn (B) Ni (C) Cu (D) Zn

36. 14.0 mL of 0.53 M barium nitrate solution and 16.0 mL of 0.44 M sodium sulfate solution are mixed. Which ion is LEAST abundant in solution?

- (A) Ba^{2+} (B) NO_3^- (C) Na^+ (D) SO_4^{2-}

37. At 120 °C and 1 atm pressure, 1.00 L of methane, CH_4 , reacts completely with excess oxygen to form carbon dioxide and water. What volumes of the two products are produced at this pressure and temperature?

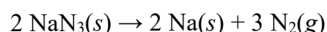
- (A) 1.00 L CO_2 and 2.00 L H_2O
(B) 1.00 L CO_2 and 4.00 L H_2O
(C) 2.00 L CO_2 and 2.00 L H_2O
(D) 2.00 L CO_2 and 4.00 L H_2O

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38. Polypropylene is made by polymerizing propene, C_3H_6 ($M = 42.1$). How many molecules of propene must be polymerized to make 3.50 g polypropylene?
- (A) 1.43×10^{22} (B) 5.01×10^{22}
(C) 6.02×10^{23} (D) 2.11×10^{24}
39. An organic compound contains only carbon, hydrogen, nitrogen, and oxygen. It is 61.71% C, 4.03% H, and 16.00% N by mass. What is its empirical formula?
- (A) C_5H_4NO (B) $C_9H_7N_2O_2$
(C) $C_{10}H_8N_2O$ (D) $C_{11}H_8NO_2$
40. 100.0 mL of 0.500 M $CaBr_2$ and 50.0 mL of 1.00 M $NaBr$ are mixed. What is the concentration of bromide ion in the resulting solution?
- (A) 0.500 M (B) 0.667 M
(C) 0.750 M (D) 1.00 M
41. 1.00 g of hydrated potassium carbonate, $K_2CO_3 \cdot nH_2O$, is heated to 250 °C to give 0.836 g anhydrous K_2CO_3 . What is the value of n ?
- (A) 0.16 (B) 1.0 (C) 1.5 (D) 2.0
42. The concentration of an aqueous solution of a nonvolatile, monoprotic acid is measured first by freezing point depression and then by boiling point elevation. The solution is found to be 0.93 m by freezing point depression and to be 0.82 m by boiling point elevation. Which is the best explanation for this discrepancy?
- (A) Ionization of the acid is markedly exothermic.
(B) The solute associates partially into dimers at lower temperatures.
(C) The volume of the solution is greater at higher temperatures.
(D) The boiling point elevation constant for water is smaller than its freezing point depression constant.
43. A 2.50 L sample of butane gas (C_4H_{10}), measured at 22.0 °C and 1.20 atm pressure, is combusted completely and the carbon dioxide gas collected at the same pressure and temperature. What volume of CO_2 is produced?
- (A) 2.50 L (B) 9.00 L
(C) 10.0 L (D) 22.5 L
44. How many hydrogen atoms are in 2.50 g of pharmacolite, $CaHAsO_4 \cdot 2H_2O$ ($M = 216.0$)?
- (A) 6.97×10^{21} (B) 2.09×10^{22}
(C) 2.79×10^{22} (D) 3.48×10^{22}
45. Methanol is produced industrially by catalytic hydrogenation of carbon monoxide according to the following equation:
- $$CO(g) + 2 H_2(g) \rightarrow CH_3OH(l)$$
- If the yield of the reaction is 40%, what volume of CO (measured at STP) would be needed to produce 1.0×10^6 kg CH_3OH ?
- (A) 2.8×10^8 L (B) 7.0×10^8 L
(C) 1.7×10^9 L (D) 2.1×10^9 L
46. What volume of 18.0 M sulfuric acid must be diluted to 250.0 mL to afford a 0.55 M solution of sulfuric acid?
- (A) 3.1 mL (B) 4.5 mL (C) 7.6 mL (D) 31 mL
47. What is the concentration of H_2O_2 in a solution that is 30.0% by mass hydrogen peroxide and has a density of 1.11 g cm^{-3} ?
- (A) 9.79 M (B) 12.6 M
(C) 18.5 M (D) 32.6 M
48. A 5.00 g mixture of potassium sulfide and potassium chloride contains 2.80 g potassium. What percentage by mass of the mixture is potassium sulfide?
- (A) 13.8% (B) 19.2% (C) 44.0% (D) 96.1%
49. Which hydrocarbon is 84.1% carbon by mass?
- (A) CH_4 (B) C_2H_6 (C) C_4H_{10} (D) C_8H_{18}
50. Lithium hydrogen carbonate, $LiHCO_3$, decomposes when heated to form Li_2O , CO_2 , and H_2O . How many moles of H_2O are formed when 0.50 mol $LiHCO_3$ decomposes?
- (A) 0.25 mol (B) 0.50 mol
(C) 0.75 mol (D) 1.0 mol
51. A vessel contains 66 g CO_2 and 16 g O_2 at a pressure of 10.0 atm. What is the partial pressure of CO_2 ?
- (A) 5.0 atm (B) 6.0 atm
(C) 7.5 atm (D) 8.0 atm

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52. First-generation automobile airbags were inflated by the decomposition of sodium azide ($M = 65.02$):



What mass of sodium azide would be required to inflate a 16.0 L airbag to a pressure of 1.20 atm at 17 °C?

- (A) 34.9 g (B) 52.4 g (C) 78.6 g (D) 157 g
53. Which of the following combinations will produce a 0.4 M NaCl solution?
- (A) Mixing 500 mL of 0.4 M NaOH with 500 mL of 0.4 M HCl
- (B) Mixing 300 mL of 1.2 M NaOH with 600 mL of 0.6 M HCl
- (C) Mixing 500 mL of 0.4 M NaCl solution with 500 mL water
- (D) Diluting 400 mL of 0.6 M NaCl with water to a final volume of 1.0 L
54. Which solute, when dissolved in 1 kg H₂O, will give the solution with the highest boiling point?
- (A) 50 g ethylene glycol, HOCH₂CH₂OH
- (B) 50 g hydroxyacetic acid, HOCH₂COOH
- (C) 50 g glycerol, HOCH₂CHOHCH₂OH
- (D) 50 g glyoxal, HCOCHO
55. How many oxygen atoms are in 225 g O₂?
- (A) 4.23×10^{24} (B) 6.84×10^{24}
- (C) 8.47×10^{24} (D) 1.69×10^{25}
56. A 0.300 M solution of HCl is prepared by adding some 1.50 M HCl to a 500 mL volumetric flask and diluting to the mark with deionized water. What volume of 1.50 M HCl must be added?
- (A) 100. mL (B) 150. mL
- (C) 225. mL (D) 250. mL
57. Copper(I) oxide, Cu₂O, is reduced to metallic copper by heating in a stream of hydrogen gas. What mass of water is produced when 10.00 g copper is formed?
- (A) 1.259 g (B) 1.417 g
- (C) 2.835 g (D) 5.670 g
58. The mineral enargite is 48.41% Cu, 19.02% As, and 32.57% S by mass. What is the empirical formula of enargite?
- (A) CuAsS (B) Cu₂AsS₂
- (C) Cu₃AsS₄ (D) Cu₄AsS₃
59. A 5.0 g sample of calcium nitrate (Ca(NO₃)₂, $M = 164$) contaminated with silica (SiO₂, $M = 60.1$) is found to contain 1.0 g calcium. What is the mass percent purity of calcium nitrate in the sample?
- (A) 20% (B) 24% (C) 73% (D) 82%
60. A solution of 20.0 g of which hydrated salt dissolved in 200 g H₂O will have the lowest freezing point?
- (A) CuSO₄ • 5 H₂O ($M = 250$)
- (B) NiSO₄ • 6 H₂O ($M = 263$)
- (C) MgSO₄ • 7 H₂O ($M = 246$)
- (D) Na₂SO₄ • 10 H₂O ($M = 286$)
61. Glyoxal consists of 41.4% C, 3.5% H, and 55.1% O by mass. What is the empirical formula of glyoxal?
- (A) CHO (B) CH₂O
- (C) CH₂O₂ (D) C₁₂HO₁₆
62. How many hydroxide ions are in 2.5 mol Mg(OH)₂?
- (A) 3.0×10^{23} (B) 6.0×10^{23}
- (C) 1.5×10^{24} (D) 3.0×10^{24}
63. In a sample consisting of 1.00 mol NaBr and 0.300 mol KI, what is the mass percent of iodine?
- (A) 24.9% (B) 32.6% (C) 47.2% (D) 83.1%
64. What is the concentration of chloride ions in a solution formed by mixing 150. mL of a 1.50 M NaCl solution with 250. mL of a 0.750 M MgCl₂ solution?
- (A) 0.563 M (B) 1.03 M
- (C) 1.50 M (D) 2.25 M

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65. Barium chloride reacts with sodium sulfate according to the following equation:
- $$\text{BaCl}_2(aq) + \text{Na}_2\text{SO}_4(aq) \rightarrow \text{BaSO}_4(s) + 2 \text{NaCl}(aq)$$
- A student mixes a solution containing 10.0 g BaCl_2 ($M = 208.2$) with a solution containing 10.0 g Na_2SO_4 ($M = 142.1$) and obtains 12.0 g BaSO_4 ($M = 233.2$). What is the percent yield of this reaction?
- (A) 60.0%
(B) 73.1%
(C) 93.3%
(D) The isolated barium sulfate is most likely wet, since the yield would otherwise be greater than 100%.
66. Which aqueous solution has the highest boiling point?
- (A) 1.0 *m* acetic acid, CH_3COOH
(B) 1.0 *m* sulfuric acid, H_2SO_4
(C) 1.0 *m* phosphoric acid, H_3PO_4
(D) 1.0 *m* glucose, $\text{C}_6\text{H}_{12}\text{O}_6$
67. A 20.0 g sample of mercury(II) oxide (HgO , $M = 216.6$) is heated strongly, causing it to decompose to metallic Hg and O_2 gas. What volume of O_2 gas is produced (measured at STP)?
- (A) 1.03 L (B) 2.07 L (C) 4.14 L (D) 14.0 L
68. When 30.0 mL of 0.10 M AgNO_3 is added to 30.0 mL of 0.10 M NaCl , aqueous NaNO_3 and solid AgCl are formed. How much solid AgCl is produced?
- (A) 0.0030 mol (B) 0.0060 mol
(C) 0.030 mol (D) 0.060 mol
69. How much $\text{Sr}(\text{OH})_2 \cdot 8 \text{H}_2\text{O}$ ($M = 265.76$) is needed to prepare 250.0 mL of solution in which $[\text{OH}^-] = 0.100 \text{ M}$?
- (A) 3.32 g (B) 6.64 g (C) 9.97 g (D) 13.3 g
70. A 10.00 g sample of a compound containing only carbon, hydrogen, and oxygen forms 23.98 g CO_2 and 4.91 g H_2O upon complete combustion. What is the empirical formula of the compound?
- (A) C_2HO (B) $\text{C}_3\text{H}_3\text{O}$ (C) $\text{C}_6\text{H}_3\text{O}_2$ (D) $\text{C}_6\text{H}_6\text{O}$
71. Which of the following is a nonelectrolyte in aqueous solution?
- (A) H_2SO_4 (B) $\text{NaC}_2\text{H}_3\text{O}_2$
(C) K_2CO_3 (D) CH_2O
72. An aqueous solution of potassium sulfate (K_2SO_4) has a freezing point of -2.24°C . What is its molality? ($K_f = 1.86^\circ\text{C}\cdot m^{-1}$)
- (A) 0.401 *m* (B) 0.602 *m*
(C) 1.20 *m* (D) 4.17 *m*